

Goal

A solution/solvent should be degassed.

Therefore, the diluted gas in the liquid and the gas of the atmosphere can either be distributed by flushing the solution with Argon or the gas can be removed out by vacuum with the freeze pump thaw method. The first one (bubbling with Argon) is preferable for high amounts of solution since a lot of solvent is evaporated during this method. The second one (Freeze pump thaw) should only be done for small amounts of liquids since high amounts increase the security issues. The idea of freeze pump is to freeze the solution with liquid N₂, so that the gas phase can be removed out using vacuum. Then, the frozen solvent is warmed, allowing the diluted gas inside the solvent to get into the atmosphere. The solution is then frozen again and the atmosphere is removed out. This is repeated three times or more. In the last cycle during the warming nearly no bubbles should be visible going out of the solution. Thus, the atmosphere is only filled with evaporated solvent. The pressure inside the vessel is the vapor pressure of the solvent. (vapor pressure is temperature dependent which means at rt the atmosphere is saturated with the solvent vapor, but in the cold there is nearly no vapor in the atmosphere. After the last warming step, the vessel can be filled with argon and activated molecular sieve 3 A should be added to dry the solvent by removing out water.

Safety Aspects: If Argon (or air) is frozen, it will evaporate fast and explosive by warming, causing the flask to break. Therefore, the stopcock of the Schlenk line should be always on vacuum and not be touched. Only the stopcock of the Schlenk should be turned on (vacuum into the Schlenk flask) or off (no vacuum into the Schlenk, only in the tubing). The flask with the frozen solvent should be warmed up under stirring in EtOH to secure an even "heat supply". A beaker with EtOH is used since this does not freeze out in contrast to water. Since Freeze Pump thaw brings the risk of imploding/exploding glass you should clean up your fume hood and you should not do other stuff at the same time. only one tube should be used of the schlenk line

Prerequisites and preparation

For the cooling you need liquid N₂, a small silver dewar

For the warming up you need: a beaker with EtOH, a stirring plate

Steps

ca. 1h

Step number	Step description
1: Put the solution Schlenk on the Schlenk line	A flame dried Schlenk flask filled with solution and a stir bar (only one half of the flask volume) is connected to the Schlenk line. The tubing (Schlauch) is flushed and set under vacuum

2. Open stop cock of the Schlenk line to vacuum	The Stop cock of the Schlenk line is set on vacuum and should <i>not be touched again</i>
3. Stir	The stirring is turned on for the whole procedure
4. Freeze	The tube is cooled down for a few minutes (ca. 5 min) in liquid N ₂ to freeze the solvent
5. Open the stop cock	The stop cock of the Schlenk is opened
6. Vacuum	The tube is set under vacuum until a pressure of $1.0 \cdot 10^{-3}$ mbar is reached
7. Close the stop cock	The stop cock of the the Schlenk is closed
8. Warm up	The nitrogen dewar is replaced with a beaker filled with EtOH to warm up the solvent (see picture)
9. Freeze	After the whole solvent is again liquified, the beaker of EtOH is removed, the flask is cleaned from the outside with a Zewa and set into the nitrogen dewar
10. Repeating Step 3-8	This procedure of cooling and warming is repeated three times
11. Fill with Ar	After the third time of warming the solvent in EtOH, the tube is filled with argon and the Schlenk is closed
12. Dry over molecular sieve 3A	Finally, molecular sieve 3A (1/3 of the solution) is filled into the tube

Attached file

Freeze-pump-thaw.jpeg

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